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Institutional Investors and Infrastructure Investing

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1 Big picture

- **Question.** Do institutional investors receive the long-term, stable, diversifying cash flows they expect from infrastructure investments?
- **Main answer.** Mostly no. Institutional investors gain exposure largely through finite-horizon closed private funds, and those funds deliver cash flows and performance closer to private equity than to stable long-duration assets.
- **Investment-structure fact.** Closed funds dominate the sample: 538 closed funds generate 59,779 investor-deal observations, far more than listed funds, open-ended funds, or direct investments.
- **Risk-return fact.** Infrastructure funds do not deliver especially stable cash flows. Capital calls and distributions are cyclical and timed similarly to buyout, VC, and real estate funds.
- **Performance fact.** Closed infrastructure funds underperform public markets and other private funds. The average PME is below one, and generalized PME is negative.
- **Exit fact.** Closed funds depend heavily on exits. A higher fraction of exited deals is strongly associated with better reported performance.
- **Investor-type fact.** Public investors allocate more to infrastructure over time and perform worse than private investors.
- **Mechanism for public investors.** ESG preferences and regulation explain 25%–40% of public investors' increased allocation to infrastructure and roughly 30% of their underperformance.
- **Implication.** Infrastructure may be socially valuable, but the common closed-fund structure does not deliver the promised long-run stable income to institutional investors.
- **Economic takeaway.** The problem is not simply the underlying asset. The contracting and investment vehicle matter. Long-lived assets packaged in finite-horizon private equity funds can inherit private-equity-like cash-flow patterns, exit pressure, and performance problems.

2 Data and measurement

2.1 Main data sources

- **Preqin infrastructure data:** investor identities, fund commitments, investment structure, underlying deals, asset sectors, and deal characteristics.
- **Burgiss cash-flow data:** fund-level cash flows used as an independent performance benchmark.
- **Sample period:** infrastructure investments from 1990 through June 2020, with most activity concentrated in 2008–2020.
- **Investor types:** public pension funds, government agencies, sovereign wealth funds, private pension funds, insurance firms/banks, and endowments/foundations.
- **Asset types:** renewable energy, traditional energy, transportation, social infrastructure, utilities, telecoms, and diversified infrastructure.

Interpretation. The paper combines investor identity, fund structure, deal characteristics, and performance. This makes it possible to ask not only whether infrastructure performs well, but also who invests, through which vehicle, and with what incentives.

2.2 Investment structures

The paper studies four main ways institutional investors obtain infrastructure exposure:

- **Closed funds:** finite-horizon private funds organized like buyout or VC funds.
- **Direct deals:** direct investor ownership positions in infrastructure assets.
- **Listed funds:** publicly traded infrastructure vehicles.
- **Open-ended funds:** evergreen vehicles without a fixed terminal date.

Interpretation. The central tension is that infrastructure is marketed as long-duration and stable, but the dominant vehicle is a closed fund that must deploy capital, manage assets, and exit deals within a finite horizon.

2.3 Performance measures

The paper uses three standard private-market performance measures:

$$PME = \frac{\text{public-market value of distributions}}{\text{public-market value of capital calls}},$$

IRR = discount rate setting net present value of fund cash flows to zero,

$$Multiple = \frac{\text{total distributions plus residual value}}{\text{paid-in capital}}.$$

Interpretation. PME is usually preferred because it compares private-fund cash flows to an investable public benchmark. IRR is sensitive to timing and scale. The multiple is intuitive but ignores the time value of money.

2.4 Generalized public market equivalent

The paper also uses the generalized PME method of Korteweg and Nagel:

$$GPME = E[m_t \cdot CF_t],$$

where m_t is a stochastic discount factor and CF_t is normalized fund cash flow.

Interpretation. GPME asks whether infrastructure cash flows earn abnormal value after allowing the discount factor to price benchmark private-market cash flows. A negative GPME means investors lose value relative to a properly risk-adjusted benchmark.

2.5 Cash-flow cyclicalities and volatility

The paper studies quarterly fund cash flows:

$$NetCF_{f,t} = Distributions_{f,t} - CapitalCalls_{f,t}.$$

It relates cash flows to market variables such as Moody's Baa–Aaa yield spread, the S&P 500 price-dividend ratio, inflation, and the Chicago Fed National Activity Index.

Interpretation. If infrastructure provided stable countercyclical cash flows, its net cash flows should be weakly related to market stress and macro conditions. The paper finds the opposite: infrastructure cash flows look cyclical.

2.6 Public versus private investor variables

- Public investor equals one for public pension funds, government agencies, and sovereign wealth funds.
- ESG preference proxies include UN PRI signatory status and impact-fund exposure.
- Regulation proxies distinguish mandatory and voluntary regulations encouraging ESG considerations.
- Performance regressions control for investor size, year of first investment, number of funds, fund type, fund and investor country fixed effects, deal geography, industry, project stage, and concession status.

Interpretation. The paper asks whether public investors perform worse because they select worse funds, face ESG/regulatory mandates, or invest with nonfinancial objectives.

2.7 Exit-rate and benchmarking data

- Deal exits are observed at the investor-deal level.
- The paper studies cumulative exit hazards across investment structures.
- It simulates cash flows from the underlying infrastructure deals using MSCI World Infrastructure index capital gains while holding exit timing fixed.
- It matches public investors' infrastructure fund investments to their buyout, VC, and real estate investments in the same vintage years.

Interpretation. The exit analysis connects the fund-performance result to the actual asset-level behavior of investment vehicles.

3 Models and empirical specifications

3.1 Deal-type regressions

For deal characteristic Y_d , the paper estimates

$$Y_d = \alpha + \beta_1 Direct_d + \beta_2 Listed_d + \beta_3 OpenEnded_d + \Gamma' X_d + \varepsilon_d.$$

Interpretation. Closed funds are the omitted category. The coefficients compare direct, listed, and open-ended investments to closed funds.

3.2 Fund performance and exits

$$Performance_f = \alpha + \beta ExitedDeals_f + \Gamma' X_f + \varepsilon_f.$$

Performance is PME, net IRR, multiple, or a reporting indicator.

Interpretation. A positive β means reported performance depends on quick exits. This is important because an asset class marketed as long-term should not require rapid exits to perform well.

3.3 Cash-flow cyclicalities

$$CF_{f,t} = \alpha_f + \delta_t + \beta_1 MarketCondition_t + \Gamma' X_{f,t} + \varepsilon_{f,t}.$$

Interpretation. If infrastructure funds truly diversify market swings, the market-condition coefficients should be small and cash flows should be smoother than other private-fund cash flows.

3.4 Public-investor allocation regressions

At the investor-vintage-year level:

$$Investments_{i,t} = \alpha + \beta Public_i + \theta ESG_{i,t} + \phi(Public_i \times ESG_{i,t}) + \Gamma' X_{i,t} + \varepsilon_{i,t}.$$

Interpretation. The interaction terms test whether public investors allocate more to infrastructure when they have ESG preferences or face ESG-related regulation.

3.5 Public-investor performance regressions

At the investor-fund level:

$$Performance_{i,f} = \alpha + \beta Public_i + \theta ESG_{i,t} + \Gamma' X_{i,f} + \varepsilon_{i,f}.$$

Interpretation. A negative public-investor coefficient means public investors underperform private investors even within the same infrastructure asset class.

3.6 Deal-exit survival model

The paper estimates Cox proportional hazard models:

$$h(t|X) = h_0(t) \exp(X'\beta).$$

Reported coefficients are hazard ratios.

Interpretation. A hazard ratio below one means lower exit probability and longer holding periods; a hazard ratio above one means faster exits.

3.7 Matched comparison with other private funds

For public investor i and vintage year t , the paper compares

$$Performance_{i,t}^{Infra} - Performance_{i,t}^{Other},$$

where other funds are buyout, VC, or real estate investments made by the same investor in the same vintage year.

Interpretation. This is a within-investor benchmark. It asks whether public investors would have performed better by committing the same capital to other private funds they were already using.

4 Identification and empirical design

4.1 What identifies the main empirical patterns

- The first set of results compares infrastructure with other private funds using fund-level performance and cash-flow data.
- The second set compares public and private investors within infrastructure, controlling for fund type, geography, industry, project stage, and concession status.
- The public-investor analysis uses investor-vintage-year and investor-fund variation, not just aggregate allocation trends.
- The matched comparison uses the same public investor and same vintage year to benchmark infrastructure against buyout, VC, and real estate investments.

Interpretation. The design is not a randomized experiment. Credibility comes from combining several complementary comparisons: cross-asset, cross-investor, within-investor, fund-level, deal-level, and cash-flow-level evidence.

4.2 Why the vehicle distinction matters

- Closed funds are finite-horizon and must exit.
- Direct deals, listed funds, and open-ended funds can hold assets longer.
- The same type of underlying asset can deliver different investor cash flows depending on how it is packaged.

Interpretation. The paper's central object is not infrastructure in the abstract; it is institutional exposure to infrastructure through specific financial structures.

4.3 Why public-investor incentives matter

- Public investors may have nonfinancial objectives, including ESG preferences and political/regulatory pressures.
- Infrastructure assets often generate positive social externalities and are linked to government priorities.
- Public investors may therefore make more marginal infrastructure investments and accept lower financial returns.

Interpretation. The paper interprets underperformance not as simple incompetence, but as partly reflecting a willingness to pay for nonfinancial objectives.

4.4 What the paper cannot fully prove

- It cannot observe all nonfinancial objectives directly.
- Preqin reporting may be selective, though Burgiss and robustness checks reduce the concern.
- The ESG and regulation proxies are imperfect.
- Matching infrastructure to other funds controls for investor and vintage year, but not for all differences in risk, mandate, or investment opportunity sets.

Interpretation. The paper is most convincing because many different tests point to the same conclusion: closed infrastructure funds behave like private equity vehicles, and public investors earn lower financial returns.

5 Tables and figures

This section keeps adjacent figures and tables together to save space and speed up reading. All visuals below are cropped directly from the paper.

5.1 Figure 1 and Table 1: investment structures and summary statistics

Read:

- Figure 1 shows that closed funds are the main channel for institutional infrastructure exposure.
- The sample includes 1,861 institutional investors and 5,907 infrastructure assets.
- Closed funds account for 538 funds and 59,779 investor-deal observations.
- Table 1 shows that public investors are important: public pension funds, government agencies, and sovereign wealth funds represent a large share of investor-deal observations.
- Most deals are secondary-stage transactions, not greenfield projects.
- Renewable energy, traditional energy, transportation, and social infrastructure are the largest sectors.

Economic message. Infrastructure is marketed as long-term real-asset exposure, but the dominant channel is a private-equity-style closed fund structure.

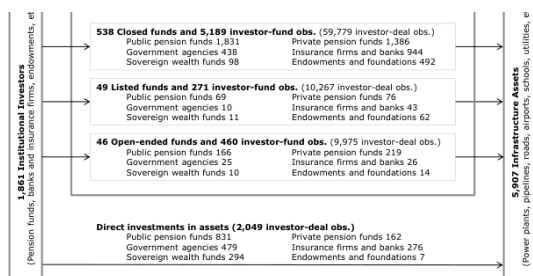


Figure 1
Investing in infrastructure
This figure depicts the investment structures through which institutional investors gain exposure to infrastructure assets. Investors can invest in infrastructure directly or through three different types of funds run by general

(a) Figure 1: investment structures

Summary statistics

	All	Public investors			Private investors		
		Public pension funds	Government agencies	Sovereign wealth funds	Private pension funds	Insurance firms and banks	Endowments and foundations
<i>A. Statistics on an institutional investor level</i>							
#Investors	1,861	409	183	37	569	338	325
Investor size	50.36	25.21	93.11	162.66	10.51	158.24	2.75
Year of first investment	2008	2008	2008	2009	2007	2008	2007
#Funds	3.27	5.04	2.56	4.15	3.12	3.05	2.03
#Direct deals	1.10	2.03	2.62	7.95	0.29	0.82	0.02
#Deals	48.16	74.76	28.38	45.84	47.54	42.92	30.34
<i>B. Distribution of investor-deal observations</i>							
#Investor-deal obs.	82,070	28,858	4,626	1,696	25,384	12,920	8,586
Secondary	60,283	21,903	2,622	1,261	19,261	9,237	5,997
Greenfield	14,862	4,762	1,569	317	3,986	2,576	1,654
Brownfield	6,925	2,193	435	118	2,137	1,107	935
No concession	73,282	25,685	3,828	1,449	22,762	11,495	8,063
With concession	8,788	3,173	798	247	2,622	1,425	523
Renewable energy	23,172	8,423	1,931	401	6,238	3,827	2,352
Traditional energy	22,940	7,088	732	427	7,091	3,554	4,048
Transport	15,043	5,026	1,070	450	5,441	2,126	930
Social	10,151	4,346	283	114	2,979	1,762	667
Utilities	6,028	2,108	267	172	2,307	798	376
Telecoms	4,450	1,756	329	123	1,234	816	192
Diversified	286	111	14	9	94	37	21
Western Europe	35,424	14,195	1,904	528	10,615	6,117	2,065
Northern America	28,435	8,990	296	386	9,073	4,455	5,235
Latin America	5,964	2,239	488	168	1,804	551	714
Asia	5,148	1,349	797	369	1,138	1,240	255
Oceania	3,904	1,149	147	135	2,066	177	230
Eastern Europe	1,896	656	390	48	518	237	47
Africa	1,299	280	604	62	170	143	40
<i>C. Statistics on an investor-deal level</i>							
#Investors in deal	1.61	1.60	1.76	1.92	1.66	1.57	1.43
%Investment stake	0.63	0.62	0.57	0.55	0.61	0.66	0.71
%Total stake	0.78	0.77	0.76	0.75	0.76	0.80	0.83

(b) Table 1: summary statistics

5.2 Table 2 and Figure 2: deal types and performance distributions

Read:

- Table 2 shows that direct deals differ from closed funds: direct investments are less likely to be greenfield or brownfield and more likely to be secondary transactions.
- Direct deals tend to involve more co-investors and smaller individual stakes.
- Figure 2 shows that infrastructure performance is not superior to buyout, VC, or real estate distributions.
- Infrastructure has relatively low PME and IRR, while the right tail is less attractive than VC and real estate.

Economic message. Direct deals appear more like minority stakes in mature assets, while closed funds resemble private equity vehicles. Infrastructure does not dominate other private fund types on performance.

Deal type and investment structure

A. Deal project stage and industry

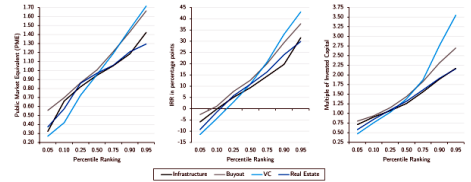
	Greenfield (1)	Brownfield (2)	Secondary (3)	Concession (4)	Renewable energy (5)	Traditional energy (6)	Transport (7)	Utilities (8)
Unconditional prob.	0.252	0.084	0.664	0.160	0.358	0.173	0.196	0.065
Direct deal	-0.050** [0.025]	-0.034*** [0.010]	0.083*** [0.029]	0.025 [0.025]	-0.115*** [0.037]	-0.031* [0.018]	0.118*** [0.025]	0.051* [0.014]
Listed	-0.111*** [0.036]	-0.052*** [0.019]	0.165*** [0.042]	-0.021 [0.035]	0.107 [0.078]	-0.125*** [0.022]	-0.019 [0.035]	-0.030* [0.014]
Open ended	-0.109*** [0.038]	-0.045*** [0.014]	0.157*** [0.042]	-0.049 [0.035]	0.106 [0.105]	-0.005 [0.035]	-0.006 [0.042]	0.034 [0.024]
Vintage FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Deal region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,665	8,649	8,665	8,651	8,591	8,652	8,651	8,594

B. Deal investment stake and size

	#Investors (1)	Investment stake (2)	Deal size (3)	Investment size (4)
Direct deal	1.151*** [0.115]	-0.299*** [0.023]	0.916*** [0.106]	0.237** [0.109]
Listed	-0.277*** [0.075]	-0.012 [0.027]	-1.098*** [0.205]	-0.830*** [0.194]
Open ended	0.052 [0.148]	-0.029 [0.053]	0.059 [0.267]	0.003 [0.243]
Vintage FE	Yes	Yes	Yes	Yes
Deal region FE	Yes	Yes	Yes	Yes
Deal industry FE	Yes	Yes	Yes	Yes
Observations	8,666	8,172	3,131	3,131
R-squared	.196	.292	.333	.270

In this table, observations are at the fund-deal level for transactions executed by infrastructure funds and investor-deal level for transactions executed directly by institutional investors. *Direct deal* is an indicator variable for direct investments in infrastructure deals. *Listed* and *Open ended* are indicators for deals accessed through

(a) Table 2: deal type and investment structure



(b) Figure 2: performance distributions

5.3 Table 3 and Figure 3: performance and cash-flow timing

Read:

- Table 3 reports average infrastructure PME of 0.93 in Prequin, below one and below the performance of buyout, VC, and real estate funds.
- Infrastructure IRR is lower than buyout, VC, and real estate in the comparable data.
- The standard deviation of infrastructure performance is not especially low.
- Figure 3 shows that capital calls and distributions over the fund life look broadly similar to other private funds.
- Infrastructure distributions are not unusually flat, stable, or longer-lived.

Economic message. Closed infrastructure funds do not deliver the long-run stable payout profile investors often expect. Their cash-flow pattern looks like other finite-horizon private funds.

Table 3
Comparison of the performance and cash flows of infrastructure funds with other private funds

	Infrastructure				Buyout				Venture capital				Real estate			
	Funds	Mean	Median	SD	Funds	Mean	Median	SD	Funds	Mean	Median	SD	Funds	Mean	Median	SD
<i>A. Fund size in \$ bil.</i>																
Preqin	315	1.35	0.72	1.86	1,811	1.37	0.50	2.52	1,765	0.30	0.15	0.55	1,663	0.60	0.30	1.09
Burgiss	201	1.58	0.82	2.03												
<i>B. Preqin performance summary statistics (equally weighted)</i>																
PME	157	0.93	0.95	0.31	1,083	1.05	1.01	0.36	839	0.99	0.95	0.56	704	0.94	0.97	0.28
IRR	210	9.92	9.35	13.53	1,534	14.52	12.70	16.46	1,419	13.33	11.00	22.52	1,477	10.58	10.90	14.71
Multiple	315	1.34	1.26	0.49	1,811	1.56	1.44	0.65	1,765	1.74	1.38	2.66	1,663	1.36	1.32	0.51
<i>C. Burgiss performance summary statistics (equally weighted)</i>																
PME	201	0.89	0.91	0.31												
IRR	201	5.60	7.94	21.55												
Multiple	201	1.23	1.20	0.45												
<i>D. Preqin performance summary statistics (value weighted)</i>																
PME	157	0.93	0.96	0.23	1,083	1.06	1.04	0.30	839	1.02	0.99	0.42	704	0.93	0.98	0.29
IRR	210	9.68	9.70	10.85	1,534	13.82	12.90	11.72	1,419	12.26	11.00	16.45	1,477	9.41	10.90	12.66
Multiple	315	1.28	1.22	0.36	1,811	1.48	1.41	0.48	1,765	1.52	1.34	0.90	1,663	1.27	1.26	0.43

We compare the performance and cash flows of closed infrastructure funds with buyout, venture capital, and real estate funds. The sample includes funds raised in the period 2002–2018. Panel A reports summary statistics on *Fund size*, which is measured in \$ bil. based on Preqin and Burgiss data. Panels B, C, and D summarize three performance measures on private funds: public market equivalent (PME) relative to the total return on the S&P 500 index, net IRR, and multiple of invested capital. We calculate the PME measure as the ratio of the sum of discounted distributions to the sum of discounted capital calls. The PME calculation uses cash flows until the end of 2019, which we downloaded from Preqin in June 2020. For each measure, we present the number of fund observations, mean, median, and standard deviation by fund type. Panel B presents equal-weighted performance measures for all fund types based on Preqin data. Panel C presents equal-weighted performance measures for infrastructure funds based on Burgiss data. Panel D presents value-weighted performance measures by fund size for all fund types based on Preqin data.

Figure 3: Table 3: performance and cash-flow comparison

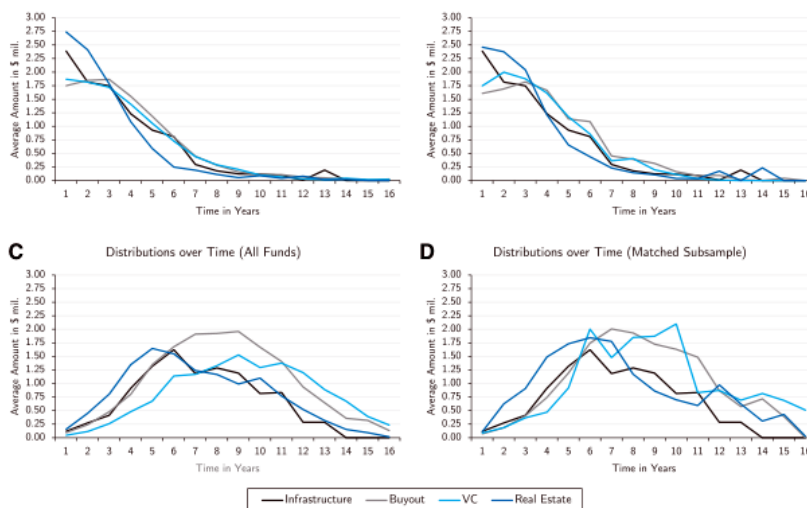


Figure 3
Comparison of the cashflows of infrastructure funds with other private funds

This figure presents the cash flows of infrastructure, buyout, venture capital (VC), and real estate funds raised in the period 2002–2018. We standardize the cash flows over time so that time period $t=1$ corresponds to the vintage year of the fund. All cash flows are based on a \$10 mil. initial commitment. We present the amounts of capital calls and distributions per year in \$ mil. In panels A and C, we use the entire sample of buyout, VC, and real estate funds. In panels B and D, we use a matched subsample of buyout, VC, and real estate funds. We create

Figure 4: Figure 3: cash flows over the fund life

5.4 Tables 4–6: exits, cash-flow cyclical, and risk-adjusted performance

Read:

- Table 4 shows that funds reporting performance are more likely to have exited deals.

- A higher share of exited deals is associated with higher PME, IRR, and multiple.
- Table 5 shows that infrastructure cash flows are related to market conditions and do not obviously hedge the business cycle.
- Diversifying infrastructure with buyout or VC does not greatly reduce cash-flow volatility.
- Table 6 shows negative GPME estimates. The full-sample GPME is negative and significant, and the value is more negative once SDF parameters adjust for market risk.

Economic message. Infrastructure funds appear to need exits to generate performance, and their cash flows do not behave like stable bond-like income. Risk-adjusted performance is weak.

Percentage exited deals and performance

	Reporting		PME		Net IRR		Multiple	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
%Exited deals	0.269***		0.636***		19.944***		0.561***	
	[0.113]		[0.206]		[3.779]		[0.191]	
%Exited deals in years 0-5		0.290*		0.851***		25.762***		0.787***
		[0.151]		[0.209]		[3.868]		[0.204]
%Exited deals in years 5-10		0.246*		0.140		9.088*		0.201
		[0.145]		[0.263]		[5.152]		[0.155]
%Exited deals in years > 10		0.201		-0.822		-8.028		-0.041
		[0.372]		[1.315]		[8.749]		[0.342]
Fund size	0.167***	0.168***	-0.014	-0.027	-2.128*	-1.785	-0.056*	-0.050
	[0.026]	[0.026]	[0.022]	[0.025]	[1.076]	[1.039]	[0.032]	[0.031]
%Greenfield	0.115	0.114	0.143	0.093	-0.261	-0.583	-0.015	-0.039
	[0.092]	[0.092]	[0.091]	[0.071]	[4.434]	[4.119]	[0.121]	[0.111]
%Brownfield	0.052	0.052	0.134	0.137	-6.643	-8.309	0.003	-0.063
	[0.125]	[0.126]	[0.168]	[0.157]	[13.056]	[13.182]	[0.226]	[0.216]
%Concession	-0.138	-0.136	-0.295	-0.285*	-13.775	-12.880	0.101	0.107
	[0.134]	[0.130]	[0.185]	[0.157]	[13.509]	[13.445]	[0.133]	[0.122]
Vintage FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
%Deal region	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
%Deal industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	526	526	125	125	183	183	260	260
Adjusted R-squared			.445	.487	.168	.198	.359	.378

In this table, observations are at the infrastructure fund level. Columns 1 and 2 present results of logit regressions in which the dependent variable equals one if a closed infrastructure fund reports the net IRR, a multiple of invested capital, or cash flows in the Preqin database. We present the marginal effects (elasticities) at the means of the independent variables. In the other columns, we limit attention to infrastructure funds reporting performance. In columns 3 and 4 performance is measured using the public market equivalent (PME), in columns 5 and 6 performance is measured using the net internal rate of return (IRR), and in columns 7 and 8 performance is measured using the net multiple of invested capital. %*Exited deals* measures the percentage of exited deals from the total number of deals made by the fund. %*Exited deals in years 0-5, 5-10, and > 10* capture the percentage of exited deals in the first 5 years after the transaction date, in 5 to 10 years after the transaction, and in more than 10 years after the transaction date, respectively. *Fund size* is the natural logarithm of the assets raised by the infrastructure fund. %*Greenfield* and %*Brownfield* measure the percentage of fund investments in deals in

(a) Table 4: exits and performance

Cash flows and business cycle

A. Infrastructure cash flows and business cycle

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ln(Yield spread)	0.342	0.320	0.722	0.550	-0.076*	-0.076*	-0.074*	-0.066*
	[0.213]	[0.219]	[0.881]	[0.813]	[0.043]	[0.043]	[0.039]	[0.038]
ln(P/D)	4.981***	4.098**			1.177***	1.140***	0.224	0.584
	[1.855]	[1.979]			[0.365]	[0.354]	[0.376]	[0.402]
Inflation		0.161				0.007		-0.065**
		[0.139]				[0.025]		[0.028]
CFNAI MA3			1.037*					
			[0.521]					
CFNAI production & income				1.530**				
				[0.734]				
Fund age FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fund focus FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,066	3,066	3,066	3,066	3,066	3,066	3,066	3,066
Adjusted R-squared	.128	.128	.127	.127				
Pseudo-R-squared					.057	.057	.090	.090

B. Diversification within fund-age and across fund-types

	Baseline buyout & buyout	Buyout & infra	Buyout & VC	Buyout & RE	Baseline VC & VC	VC & infra	VC & buyout	VC & RE
SD baseline all funds	4.825	4.984	4.752	4.897	5.867	5.962	5.837	5.951
SD baseline funds above median	5.184	5.443	5.107	5.346	6.425	6.591	6.400	6.566
SD baseline funds above 75 th perc.	5.631	5.941	5.536	5.864	7.491	7.764	7.378	7.806

In panel A, we examine the relation between cash flows of infrastructure funds and market conditions. The unit of observation is by fund-quarter. *ln(P/D)* is the natural logarithm of the price/dividend ratio of the S&P 500. *ln(Yield spread)* is the natural logarithm of the Moody's Baa-Aaa yield spread. We orthogonalize *ln(Yield spread)* with respect to *ln(P/D)*. *Inflation* is the U.S. CPI for all urban consumers. *CFNAI MA3* is the 3-month moving average of the Chicago Fed National Activity index. The *CFNAI Production & Income* category of this index measures industrial production, manufacturing, construction, and real income. All variables are lagged and

(b) Table 5: cash flows and business cycle

Generalized public market equivalent

	All funds		Size ≥ \$250 mil.		Size ≥ \$500 mil.	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>A. Funds raised in 2002–2018</i>						
GPME	-0.067**	-0.257**	-0.069**	-0.195	-0.095***	-0.284***
	(.013)	(.015)	(.013)	(.208)	(.001)	(.004)
a	0	0.111	0	0.111	0	0.096
		[0.032]		[0.035]		[0.034]
b	1	6.269	1	6.168	1	5.749
		[1.099]		[1.156]		[1.188]
Funds	145	145	121	121	100	100
<i>B. Funds raised in 2002–2013</i>						
GPME	-0.100**	-0.338***	-0.091**	-0.302**	-0.138***	-0.383***
	(.014)	(.006)	(.038)	(.031)	(.001)	(.000)
a	0	0.079	0	0.076	0	0.063
		[0.025]		[0.026]		[0.025]
b	1	5.109	1	4.999	1	4.577
		[0.903]		[0.956]		[0.974]
Funds	67	67	56	56	47	47

In this table, we use the stochastic discount factor (SDF) valuation method of Korteweg and Nagel (2016) to estimate the generalized public market equivalent (GPME). The unit of observation is infrastructure fund cash flows on a quarterly frequency. The quarterly cash flows are normalized by fund size to a total commitment of \$1, and they are calculated as the difference between distributions and capital calls, rather than their ratio (so a PME equal to one in Table 3 corresponds to a GPME equal to zero in this table). In panel A, we analyze the cash flows of funds raised in the 2002–2018 period, while in panel B, we analyze the cash flows funds raised in the

Figure 6: Table 6: generalized PME

5.5 Figure 4 and Table 7: rising AUM and public-investor allocation

Read:

- Figure 4 shows infrastructure closed-fund AUM rising from about \$59 billion in 2008 to about \$486 billion in 2019.
- The public-investor share of investor-deal observations rises from 37% in 2008 to 57% in 2019.
- Table 7 shows that public investors make more infrastructure investments.
- UN PRI signatory status and mandatory regulation are positively related to infrastructure allocations.
- Interactions indicate that ESG preferences and regulation are especially relevant for public investors.

Economic message. The growth of infrastructure as an asset class is driven substantially by public investors, and part of that growth is tied to ESG preferences and regulation.

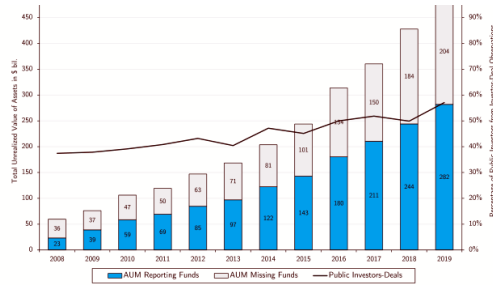


Figure 4
Assets under management and percentage of public investors
In this figure, we rely on the reported unrealized value of assets to estimate the amount of assets under management (AUM) in \$ bil. by closed infrastructure funds. Specifically, we download the time-series of annual performance

(a) Figure 4: AUM and public-investor share

	Number of investments				ln(Number of investments + 1)			
	1990-2020	2011-2020	1990-2020	2011-2020	1990-2020	2011-2020	1990-2020	2011-2020
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Public investor	0.124*** [0.016]	0.072*** [0.011]	0.311*** [0.038]	0.203*** [0.032]	0.055*** [0.006]	0.038*** [0.005]	0.133*** [0.013]	0.109*** [0.013]
UN PRI signatory	0.385*** [0.086]	0.038 [0.037]	0.343*** [0.083]	0.045 [0.045]	0.138*** [0.025]	0.031* [0.018]	0.121*** [0.024]	0.038** [0.019]
Mandatory regulation	0.151* [0.083]	0.057 [0.037]	0.398** [0.158]	0.204** [0.093]	0.035* [0.019]	0.026 [0.016]	0.077** [0.033]	0.085*** [0.029]
Voluntary regulation	0.014 [0.026]	-0.007 [0.019]	0.093 [0.063]	0.074 [0.067]	0.011 [0.011]	-0.002 [0.009]	0.046** [0.019]	0.039** [0.020]
Public investor × UN PRI signatory		0.759*** [0.178]		0.692*** [0.191]		0.240*** [0.051]		0.198*** [0.052]
Public investor × Mandatory regulation		0.162 [0.183]		0.127 [0.240]		-0.004 [0.043]		-0.052 [0.053]
Public investor × Voluntary regulation		0.002* [0.054]		0.065 [0.086]		0.047** [0.021]		0.019 [0.030]
log investor size	0.037*** [0.004]	0.036*** [0.004]	0.074*** [0.009]	0.073*** [0.009]	0.017*** [0.001]	0.017*** [0.001]	0.032*** [0.003]	0.032*** [0.003]
Year of first investment	-0.010*** [0.001]	-0.009*** [0.001]	-0.009*** [0.002]	-0.008*** [0.002]	-0.005*** [0.000]	-0.005*** [0.000]	-0.002*** [0.001]	-0.002*** [0.001]
LP country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Vintage FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	57,691	57,691	18,610	18,610	57,691	57,691	18,610	18,610
Adjusted R-squared	.091	.103	.087	.097	.136	.144	.128	.135

This table analyzes the number of infrastructure investments per year by institutional investors. Observations are at the investor-vintage year level. If an institutional investor did not make any investments in a certain year, we replace the observation with zero. In columns 1 to 4, the dependent variable is the number of investments in infrastructure funds by investor-vintage year. In columns 5 to 8, the dependent variable is the natural logarithm of the number of investments plus one. We analyze the number of investments per year separately over the entire sample 1990-2020 and over the 2011-2020 period. *Public investor* is an indicator variable for institutional investors from the public sector. *UN PRI signatory* is an indicator variable for investors that have signed the UN principles for responsible investing. It varies by investor-year and equals to one in the year of the signing date and in the following period. *Mandatory regulation* and *Voluntary regulation* are indicator variables measuring whether an institutional investor faces mandatory or voluntary regulation to consider ESG factors in the investment decisions. The regulation variables also vary by investor-year, and they equal to one in the years after the mandatory or voluntary regulatory policy is enacted and zero otherwise. Estimations in columns 2, 4, 6 and 8 include also interaction terms between public investors and *UN PRI signatory*, *Mandatory regulation*, and *Voluntary regulation*. We control for the natural logarithm of investors' size (AUM) and year of first infrastructure investment. All estimations include vintage year fixed effects and investor (LP) country fixed effects. We cluster standard errors by institutional investor and report standard errors in brackets. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 7: number of investments

5.6 Tables 8–9: public investor underperformance and explanations

Read:

- Table 8 shows that public investors perform worse in infrastructure funds.
- The public-investor coefficient is negative across PME, net IRR, and multiple.
- U.S. public pension funds, non-U.S. public pension funds, and sovereign wealth funds are especially weak performers in several specifications.
- Table 9 adds regulation, UN PRI signatory status, and impact funds.
- ESG-related variables explain part of the public-investor underperformance, but not all of it.
- Impact funds are associated with lower financial returns, consistent with nonfinancial objectives.

Economic message. Public investors appear willing to accept lower financial returns in infrastructure, partly because of ESG, regulatory, or social-externality objectives.

Investor type and performance

	PME		Net IRR		Multiple	
	(1)	(2)	(3)	(4)	(5)	(6)
Public investor	-0.026** [0.012]		-1.810*** [0.588]		-0.038*** [0.014]	
U.S. public pension fund		-0.026* [0.016]		-1.978** [0.962]		-0.024 [0.021]
Non-U.S. public pension fund		-0.024 [0.019]		-1.489* [0.764]		-0.050** [0.020]
Government agency		-0.001 [0.035]		-1.057 [1.491]		-0.082 [0.052]
Sovereign wealth fund		-0.069** [0.028]		-4.196** [2.022]		-0.052 [0.039]
log investor size	0.006** [0.003]	0.006** [0.002]	-0.018 [0.117]	0.010 [0.112]	-0.001 [0.004]	-0.001 [0.004]
Year of first investment	-0.001* [0.001]	-0.001 [0.001]	-0.044 [0.033]	-0.031 [0.040]	-0.001 [0.001]	-0.001 [0.001]
#Funds	-0.000 [0.001]	-0.000 [0.001]	-0.005 [0.026]	-0.003 [0.025]	0.000 [0.001]	0.000 [0.001]
Fund type	Yes	Yes	Yes	Yes	Yes	Yes
LP country FE	Yes	Yes	Yes	Yes	Yes	Yes
Vintage FE	Yes	Yes	Yes	Yes	Yes	Yes
%Deal region	Yes	Yes	Yes	Yes	Yes	Yes
%Deal industry	Yes	Yes	Yes	Yes	Yes	Yes
%Project stage	Yes	Yes	Yes	Yes	Yes	Yes
%Concession	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,342	2,342	3,021	3,021	3,853	3,853
Adjusted R-squared	.450	.450	.150	.150	.326	.326

This table presents results of regressions in which the dependent variable is the performance of investors in closed infrastructure funds. Observations are at the investor-fund level. In columns 1 and 2 performance is measured using the public market equivalent (PME), in columns 3 and 4 using the net internal rate of return (IRR), and in columns 5 and 6 using the net multiple of invested capital. *Public investor* is an indicator variable for institutional investors from the public sector. We also split the public investors by type and include separate indicator variables for U.S. public pension funds, non-U.S. public pension funds, government agencies, and sovereign wealth funds. We control for the natural logarithm of investors' size (AUM) and year of first infrastructure investment. *#Funds* measures the number of investments in infrastructure funds by investor. We include two indicator variables for infrastructure funds that do not take only equity positions in infrastructure deals, but that also act as a fund-of-funds or debt fund. We include vintage year fixed effects and investor (LP) country fixed effects. We also control

Figure 8: Table 8: investor type and performance

Table 9
Explanations of public investor underperformance

	PME			Net IRR			Multiple		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Public investor	-0.026** [0.012]	-0.020* [0.012]	-0.019 [0.012]	-1.823*** [0.594]	-1.475** [0.580]	-1.497** [0.583]	-0.038*** [0.015]	-0.025* [0.014]	-0.025* [0.014]
Mandatory regulation	0.010 [0.036]		0.018 [0.037]	-2.676 [1.802]		-2.485 [1.883]	-0.068* [0.037]		-0.057 [0.036]
Voluntary regulation	-0.001 [0.033]		-0.022 [0.035]	0.993 [1.432]		-0.448 [1.420]	-0.011 [0.060]		-0.042 [0.058]
UN PRI signatory		-0.027 [0.018]	-0.028 [0.018]		-1.459* [0.782]	-1.346* [0.789]		-0.042* [0.022]	-0.039* [0.023]
Impact fund		-0.175* [0.097]	-0.176* [0.097]		-11.453*** [3.508]	-11.466*** [3.531]		-0.363*** [0.100]	-0.365*** [0.100]
log investor size	0.006** [0.003]	0.005** [0.002]	0.005** [0.002]	-0.008 [0.118]	-0.005 [0.103]	-0.011 [0.103]	-0.001 [0.004]	0.000 [0.004]	-0.001 [0.004]
Year of first investment	-0.001* [0.001]	-0.001* [0.001]	-0.001* [0.001]	-0.043 [0.033]	-0.025 [0.027]	-0.025 [0.027]	-0.001 [0.001]	-0.000 [0.001]	-0.000 [0.001]
#Funds	-0.000 [0.001]	0.000 [0.001]	0.000 [0.001]	-0.006 [0.030]	0.003 [0.028]	0.003 [0.029]	0.000 [0.001]	0.000 [0.001]	0.000 [0.001]
Fund type	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
LP country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Vintage FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
%Deal region	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
%Deal industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
%Project stage	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
%Concession	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,342	2,342	2,342	3,021	3,021	3,021	3,853	3,853	3,853
Adjusted R-squared	.450	.469	.469	.151	.199	.200	.326	.370	.370

This table presents results of regressions in which the dependent variable is the performance of investors in closed infrastructure funds. Observations are at the investor-fund level. In columns 1 to 3, performance is measured using the public market equivalent (PME), in columns 4 to 6 using the net internal rate of return (IRR), and in columns 7 to 9 using the net multiple of invested capital. *Public investor* is an indicator variable for institutional investors from the public sector. *Mandatory regulation* and *Voluntary regulation* are indicator variables measuring whether an institutional investor faces mandatory or voluntary regulation to consider ESG factors in its investment decisions. *UN PRI signatory* is an indicator variable for investors that have signed the UN principles for responsible investing. *Impact fund* is an indicator for infrastructure funds that make investments with the intention to generate positive social and environmental impact alongside a financial return. We control for the natural logarithm of investors' size (AUM) and year of first infrastructure investment. *#Funds* measures the number of investments in infrastructure funds by investor. We include two indicator variables for infrastructure funds that do not take only equity positions in infrastructure deals, but that also act as a fund-of-funds or debt fund. We include vintage year fixed effects and investor (LP) country fixed effects. We also control for the percentage of deals in the portfolio of each infrastructure fund in different industries, geographical regions, project stages, and deals backed with a concession agreement. We double cluster standard errors by institutional investor and infrastructure fund, and report standard errors in brackets. * $p < .1$; ** $p < .05$; *** $p < .01$.

Figure 9: Table 9: explanations of public investor underperformance

5.7 Figure 5 and Tables 10–11: exits and matched benchmarks

Read:

- Figure 5 shows that closed funds exit infrastructure assets faster than direct, listed, or open-ended vehicles.
- Simulated cash flows show that closed-fund cash flows rely heavily on asset sales and declining remaining NAV.
- Table 10 confirms that direct deals, listed funds, and open-ended funds have lower exit hazards than closed funds.
- Public investors also have lower exit hazards and hold deals longer.
- Table 11 compares public investors' infrastructure investments with their own buyout, VC, and real estate investments in the same vintage years.
- Infrastructure underperforms buyout by 5.78 IRR points and 0.21 multiple points, underperforms VC by 5.46 IRR points and 0.39 multiple points, and underperforms real estate by 1.73 IRR points.

Economic message. The finite-horizon closed-fund structure forces infrastructure into a quick-exit model. Public investors would generally have earned more in other private funds they were already using.

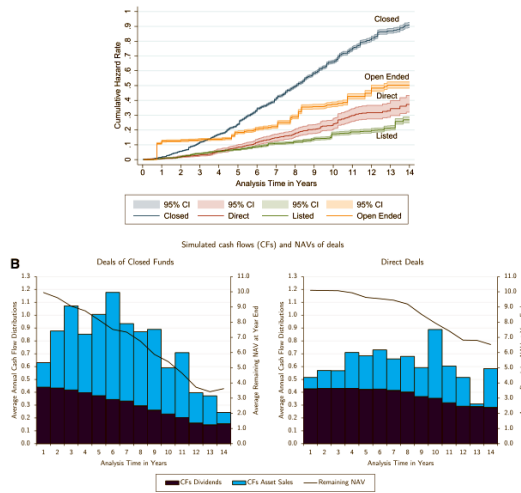


Figure 5

(a) Figure 5: exits and simulated cash flows

	(1)	(2)	(3)	(4)
Direct deal	0.491*** [0.077]	0.478*** [0.074]	0.504*** [0.078]	0.490*** [0.076]
Listed	0.291*** [0.023]	0.307*** [0.023]	0.296*** [0.024]	0.311*** [0.024]
Open ended	0.698*** [0.033]	0.699*** [0.033]	0.700*** [0.033]	0.699*** [0.033]
Public investor	0.926** [0.033]	0.936* [0.032]	0.927** [0.033]	0.938* [0.032]
UN PRI signatory		0.972 [0.047]		0.973 [0.047]
Impact fund		0.551*** [0.036]		0.558*** [0.036]
Mandatory regulation			0.630*** [0.080]	0.666*** [0.079]
Voluntary regulation				1.041 [0.089]
log investor size	0.999 [0.009]	1.001 [0.009]	0.999 [0.009]	1.000 [0.009]
Year of first investment	0.984*** [0.003]	0.984*** [0.003]	0.985*** [0.003]	0.985*** [0.003]
#Funds	0.998 [0.002]	0.998 [0.002]	0.998 [0.002]	0.998 [0.002]
Concession	1.650*** [0.071]	1.650*** [0.071]	1.643*** [0.071]	1.642*** [0.070]
Greenfield	0.711*** [0.022]	0.721*** [0.022]	0.713*** [0.023]	0.722*** [0.022]
Brownfield	0.664*** [0.022]	0.662*** [0.022]	0.663*** [0.022]	0.661*** [0.022]
Home deal	0.701*** [0.042]	0.695*** [0.040]	0.704*** [0.043]	0.696*** [0.041]
#Investors	0.956*** [0.007]	0.956*** [0.007]	0.956*** [0.007]	0.956*** [0.007]
Investment stake	0.987 [0.042]	1.032 [0.045]	0.992 [0.042]	1.035 [0.046]
LP country FE	Yes	Yes	Yes	Yes
Deal region FE	Yes	Yes	Yes	Yes
Deal industry FE	Yes	Yes	Yes	Yes
Observations	79,705	79,705	79,705	79,705

This table presents results of a survival analysis using the Cox proportional hazard model. The event of interest is a sale transaction that results in a full (not partial) exit of an equity position in an asset. We present the hazard ratios. *Direct deal* is an indicator variable for direct investments in infrastructure deals. *Listed* and *Open ended* are indicators for deals accessed through listed and open-ended funds. The omitted investment structure is closed funds. *Public investor* is an indicator variable for institutional investors from the public sector. The FSG

(b) Table 10: exit hazards

Comparison of investments in infrastructure with investments in other funds

	Net IRR				Multiple			
	Obs.	Infra	Other	Diff.	Obs.	Infra	Other	Diff.
Infrastructure vs. buyout	719	8.53	14.31	−5.78***	933	1.29	1.50	−0.21***
Infrastructure vs. VC	427	8.53	13.99	−5.46***	555	1.29	1.66	−0.39***
Infrastructure vs. real estate	516	8.45	10.18	−1.73**	638	1.28	1.29	−0.01

This table compares the performance of investments in infrastructure funds with the investments in buyout, venture capital (VC), and real estate funds made by the same public institutional investor. We match the infrastructure investments with buyout, VC, and real estate investments made by the same public investor and in the same vintage year. We present the average net IRR and multiple of invested capital delivered by infrastructure, buyout, VC, or real estate funds. The number of observations changes because it depends on making investments in infrastructure, buyout, VC, or real estate in the same vintage year and on the availability of performance data for these investments. In the *Diff.* columns, we present the difference between the performance that public investors

Figure 11: Table 11: infrastructure versus other private funds

6 Short summary

- Institutional investors expect infrastructure to provide stable, long-duration, inflation-linked, and diversifying cash flows.
- The main exposure channel is not a long-horizon vehicle but a finite-horizon closed private fund.
- Closed funds dominate infrastructure exposure in the data.
- Infrastructure closed funds have low PME, lower IRR than comparable private-fund categories, and no obvious cash-flow stability advantage.
- Reported performance depends strongly on exits.
- Infrastructure cash flows are related to market conditions and business-cycle variables.
- GPME estimates imply weak risk-adjusted performance.
- Infrastructure AUM grows sharply from 2008 to 2019 despite weak performance.
- Public investors drive much of the growth in infrastructure allocations.
- Public investors perform worse than private investors.
- ESG preferences and regulation explain a substantial share of public investor allocation growth and roughly 30% of underperformance.
- Direct, listed, and open-ended structures hold assets longer than closed funds.
- Public investors' infrastructure fund investments underperform their own buyout, VC, and real estate fund investments.
- The central lesson is that asset characteristics do not automatically translate into investor cash-flow characteristics when the asset is packaged through the wrong financial vehicle.

7 Conclusion

7.1 Core conclusion

Andonov, Kräussl, and Rauh show that institutional infrastructure investing does not generally deliver the stable, long-duration, low-correlation cash flows promised by the asset-class narrative. The key reason is that institutional exposure is dominated by closed funds, whose finite-horizon structure pushes infrastructure assets into a private-equity-style model of capital calls, exits, and performance reporting.

7.2 Economic intuition

Infrastructure assets may be long-lived and socially valuable, but a closed fund has to raise capital, deploy it, manage assets, and exit within a limited window. This creates pressure to realize gains through sales rather than provide long-run distributions. The vehicle transforms the economics of the asset.

7.3 Takeaway

The paper is a warning against evaluating infrastructure only from the perspective of physical asset characteristics. What matters to investors is the full contracting structure: investor type, fund horizon, exit incentives, reporting, ESG objectives, and regulation. Infrastructure can be valuable, but closed-fund infrastructure is not automatically a stable long-term asset class.